K Aishwarya Chetana Major Project :

 # -- coding: utf-8 -*-*

"""aishmajoredi (1).ipynb

Automatically generated by Colab.

Original file is located at

https://colab.research.google.com/drive/1nic-mn8Ma82uW1tTpvaQwuCnhfKt-3sn?usp=sharing

"""

// the deploy model streamlit link ***: https://heart-disease-prediction-app-ixkgsxzj5yfy5qnnophjcq.streamlit.app/***

// Also use gradio in last cell, u can check it.

 # Standard imports for analysis and modelling

```
import os

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV, RandomizedSearchCV, cross_val_score

from sklearn.preprocessing import StandardScaler, OneHotEncoder

from sklearn.compose import ColumnTransformer

from sklearn.pipeline import Pipeline

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score, confusion_matrix, classification_report, roc_curve, auc

from sklearn.linear_model import LogisticRegression

from sklearn.tree import DecisionTreeClassifier

from sklearn.ensemble import RandomForestClassifier

import joblib
```

```
import tensorflow as tf

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense, Dropout

from tensorflow.keras.callbacks import EarlyStopping

import warnings

warnings.filterwarnings("ignore")


sns.set(style='whitegrid')

RSEED = 42

print("Imports complete.")


# Load the dataset. Change file_path if needed (e.g., '/content/drive/MyDrive/heart.csv' after mounting Drive)

file_path = "heart.csv"

if not os.path.exists(file_path):

    raise FileNotFoundError(f"{file_path} not found. Upload 'heart.csv' to Colab or change file_path.")


df = pd.read_csv(file_path)

print("Dataset loaded. Shape:", df.shape)

print("\nFirst 5 rows:")

display(df.head())

print("\nData types:")

print(df.dtypes)

print("\nMissing values per column:")

print(df.isnull().sum())


print("Column names and quick meanings:")

cols = list(df.columns)

for c in cols:
```

```
    print("-", c)

print("\nTarget value counts (0 = no disease, 1 = disease):")
print(df['target'].value_counts())

print("Summary statistics (numeric):")
display(df.describe().T)

print("\nObservations to note:")
print("- Check min/max for suspicious values (e.g., 'thal' or 'ca' sometimes encoded with 0 or missing).")
print("- Dataset rows:", df.shape[0])

# Histograms for numeric columns
num_cols = df.select_dtypes(include=[np.number]).columns.tolist()
if 'target' in num_cols:
    num_cols.remove('target')

plt.figure(figsize=(14,10))
df[num_cols].hist(bins=15, layout=(4,4), figsize=(14,12))
plt.suptitle("Histograms of numeric features")
plt.show()

cat_cols = ['sex', 'cp', 'fbs', 'restecg', 'exang', 'slope', 'ca', 'thal']
print("Categorical-like columns used for plotting:", cat_cols)

for c in cat_cols:
    if c in df.columns:
        print(f"\nValue counts for {c}:")
        print(df[c].value_counts())
```

```
    plt.figure(figsize=(6,3))
    sns.countplot(data=df, x=c, hue='target')
    plt.title(f"{c} distribution by target")
    plt.show()

plt.figure(figsize=(12,10))
corr = df.corr()
sns.heatmap(corr, annot=True, fmt=".2f", cmap='coolwarm', square=True)
plt.title("Correlation matrix")
plt.show()

print("Look for features strongly correlated with 'target' and potential multicollinearity.")

print("Checking for any non-numeric or unusual encodings:")
for col in df.columns:
  if df[col].dtype == object:
    print(col, df[col].unique()[:20])

print("\nCheck for zeros or impossible values in columns that should be positive (like 'thal' sometimes 0):")
for c in ['thal','ca']:
  if c in df.columns:
    print(c, "unique values:", sorted(df[c].unique()))

print("\nIf any '?' or string placeholders are present, we should replace them and convert columns to numeric. This dataset looks numeric-clean.")

# Prepare X and y
X = df.drop(columns=['target'])
```

```
y = df['target']

# Choose columns to one-hot encode (cp, thal, slope) as per project plan
onehot_cols = [c for c in ['cp','thal','slope'] if c in X.columns]
# Numerical columns = all remaining columns
num_cols = [c for c in X.columns if c not in onehot_cols]

print("One-hot columns:", onehot_cols)
print("Numerical columns:", num_cols)

# ColumnTransformer: scale numeric, one-hot encode chosen categorical
preprocessor = ColumnTransformer(transformers=[
    ('num', StandardScaler(), num_cols),
    ('ohe', OneHotEncoder(drop='first', sparse_output=False), onehot_cols)
], remainder='passthrough')  # passthrough in case any others remain (shouldn't)

# Fit the preprocessor and transform to inspect shape
X_pre = preprocessor.fit_transform(X)
print("Preprocessing done. Transformed shape:", X_pre.shape)

# Build final feature names list for interpretation
ohe = preprocessor.named_transformers_['ohe']
ohe_cols = list(ohe.get_feature_names_out(onehot_cols)) if len(onehot_cols)>0 else []
feature_names = num_cols + ohe_cols
print("Sample feature names (first 20):", feature_names[:20])

X_train, X_test, y_train, y_test = train_test_split(X_pre, y, test_size=0.2, random_state=RSEED, stratify=y)
print("Train shape:", X_train.shape, "Test shape:", X_test.shape)
print("Train target distribution:\n", np.bincount(y_train))
```

```
print("Test target distribution:\n", np.bincount(y_test))


def evaluate_model(name, model, X_train, y_train, X_test, y_test, show_confusion=True):
    print(f"\n>>> Evaluating: {name}")
    # prediction
    y_pred = None
    y_proba = None

    if hasattr(model, "predict_proba"):
        y_proba = model.predict_proba(X_test)[:,1]
        y_pred = (y_proba > 0.5).astype(int)
    elif isinstance(model, tf.keras.Model): # Specifically handle Keras models
        y_proba = np.ravel(model.predict(X_test))
        y_pred = (y_proba > 0.5).astype(int)
    else:
        y_pred = model.predict(X_test)
        # If it's a model that doesn't have predict_proba but predict gives scores/continuous output, try to binarize
        if y_pred.dtype == float or y_pred.dtype == np.float32 or y_pred.dtype == np.float64:
            y_proba = np.ravel(y_pred)
            y_pred = (y_proba > 0.5).astype(int)

    # Ensure y_pred is binarized for metrics
    if y_pred is None or not (np.isin(y_pred, [0, 1]).all()): # If prediction is not already binary
        raise ValueError("Model prediction output could not be binarized for evaluation.")

    # metrics
    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred)
```

```
rec = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
# roc
roc = None
try:
    # Reuse y_proba if already calculated, otherwise calculate it
    if y_proba is not None:
        roc = roc_auc_score(y_test, y_proba)
    elif hasattr(model, "predict_proba"):
        roc = roc_auc_score(y_test, model.predict_proba(X_test)[:,1])
    else: # For models like Keras where predict returns probabilities
        roc = roc_auc_score(y_test, np.ravel(model.predict(X_test)))
except Exception: # Fallback if ROC calculation fails for any reason
    roc = None

print("Accuracy:", round(acc,4))
print("Precision:", round(prec,4))
print("Recall:", round(rec,4))
print("F1-score:", round(f1,4))
print("ROC-AUC:", round(roc,4) if roc is not None else "N/A")
print("\nClassification report:")
print(classification_report(y_test, y_pred))
if show_confusion:
    cm = confusion_matrix(y_test, y_pred)
    plt.figure(figsize=(4,3))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
    plt.title(f"Confusion matrix: {name}")
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
```

```
    plt.show()
  return {'accuracy':acc, 'precision':prec, 'recall':rec, 'f1':f1, 'roc':roc}

# Logistic Regression
print("Training Logistic Regression...")
lr = LogisticRegression(max_iter=1000, random_state=RSEED)
lr.fit(X_train, y_train)
lr_metrics = evaluate_model("Logistic Regression", lr, X_train, y_train, X_test, y_test)

# Decision Tree
print("Training Decision Tree...")
dt = DecisionTreeClassifier(random_state=RSEED)
dt.fit(X_train, y_train)
dt_metrics = evaluate_model("Decision Tree", dt, X_train, y_train, X_test, y_test)

# Random Forest baseline
print("Training Random Forest (100 trees)...")
rf = RandomForestClassifier(n_estimators=100, random_state=RSEED)
rf.fit(X_train, y_train)
rf_metrics = evaluate_model("Random Forest (base)", rf, X_train, y_train, X_test, y_test)

print("Hyperparameter tuning (RandomizedSearchCV) for RandomForest — may take a few minutes.")
param_dist = {
  'n_estimators': [50,100,200,300],
  'max_depth': [None,5,10,20],
  'min_samples_split': [2,5,10],
  'min_samples_leaf': [1,2,4],
  'max_features': ['sqrt','log2', None]
}
```

```
rf_clf = RandomForestClassifier(random_state=RSEED)

rand_search = RandomizedSearchCV(rf_clf, param_distributions=param_dist, n_iter=20,
scoring='roc_auc', cv=5, random_state=RSEED, n_jobs=-1, verbose=1)

rand_search.fit(X_train, y_train)

print("Best params:", rand_search.best_params_)

best_rf = rand_search.best_estimator_

rf_tuned_metrics = evaluate_model("Random Forest (tuned)", best_rf, X_train, y_train, X_test, y_test)


print("Training a simple feedforward neural network (Keras).")


input_dim = X_train.shape[1]


def build_simple_nn(input_dim):

  model = Sequential()

  model.add(Dense(64, activation='relu', input_shape=(input_dim,)))

  model.add(Dropout(0.2))

  model.add(Dense(32, activation='relu'))

  model.add(Dropout(0.1))

  model.add(Dense(1, activation='sigmoid'))

  model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

  return model


nn_model = build_simple_nn(input_dim)

nn_model.summary()


es = EarlyStopping(monitor='val_loss', patience=15, restore_best_weights=True, verbose=1)

# Keras expects arrays

history = nn_model.fit(X_train, y_train, validation_split=0.2, epochs=200, batch_size=16, callbacks=[es],
verbose=1)
```

```
# Evaluate using our function
nn_metrics = evaluate_model("Neural Network", nn_model, X_train, y_train, X_test, y_test)


# Plot training history
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plt.plot(history.history['loss'], label='train_loss')
plt.plot(history.history['val_loss'], label='val_loss')
plt.title('Loss')
plt.legend()
plt.subplot(1,2,2)
plt.plot(history.history['accuracy'], label='train_acc')
plt.plot(history.history['val_accuracy'], label='val_acc')
plt.title('Accuracy')
plt.legend()
plt.show()


results = pd.DataFrame({
    'model': ['Logistic Regression', 'Decision Tree', 'Random Forest (base)', 'Random Forest (tuned)', 'Neural Network'],
    'accuracy': [lr_metrics['accuracy'], dt_metrics['accuracy'], rf_metrics['accuracy'], rf_tuned_metrics['accuracy'], nn_metrics['accuracy']],
    'precision': [lr_metrics['precision'], dt_metrics['precision'], rf_metrics['precision'], rf_tuned_metrics['precision'], nn_metrics['precision']],
    'recall': [lr_metrics['recall'], dt_metrics['recall'], rf_metrics['recall'], rf_tuned_metrics['recall'], nn_metrics['recall']],
    'f1': [lr_metrics['f1'], dt_metrics['f1'], rf_metrics['f1'], rf_tuned_metrics['f1'], nn_metrics['f1']],
    'roc_auc': [lr_metrics['roc'], dt_metrics['roc'], rf_metrics['roc'], rf_tuned_metrics['roc'], nn_metrics['roc']]
```

```
})
print("Model comparison (sorted by ROC-AUC):")
display(results.sort_values(by='roc_auc', ascending=False).round(4))

# RandomForest feature importances
try:
    importances = best_rf.feature_importances_
    fi = pd.Series(importances, index=feature_names).sort_values(ascending=False)
    print("Top 15 features by RandomForest importance:")
    display(fi.head(15))
    plt.figure(figsize=(8,6))
    fi.head(15).plot(kind='barh')
    plt.gca().invert_yaxis()
    plt.title("Top 15 Feature Importances")
    plt.show()
except Exception as e:
    print("Could not extract feature importances from RandomForest:", e)

# Logistic regression coefficients (linear importance)
try:
    coefs = lr.coef_.ravel()
    coef_ser = pd.Series(coefs, index=feature_names).sort_values(key=lambda x: np.abs(x),
ascending=False)
    print("\nTop features by absolute LogisticRegression coefficient:")
    display(coef_ser.head(15))
except Exception as e:
    print("LogisticRegression coefficients extraction failed:", e)

plt.figure(figsize=(8,6))
```

```
def get_proba(model, X):
    if hasattr(model, "predict_proba"):
        return model.predict_proba(X)[:,1]
    else:
        return np.ravel(model.predict(X))

models_for_roc = {
    'Logistic Regression': lr,
    'Random Forest (tuned)': best_rf,
}

for name, model in models_for_roc.items():
    y_proba = get_proba(model, X_test)
    fpr, tpr, _ = roc_curve(y_test, y_proba)
    plt.plot(fpr, tpr, label=f"{name} (AUC = {roc_auc_score(y_test, y_proba):.3f})")

# NN
try:
    y_proba_nn = np.ravel(nn_model.predict(X_test))
    fpr, tpr, _ = roc_curve(y_test, y_proba_nn)
    plt.plot(fpr, tpr, label=f"Neural Network (AUC = {roc_auc_score(y_test, y_proba_nn):.3f})")
except Exception as e:
    print("NN ROC error:", e)

plt.plot([0,1],[0,1],'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curves')
```

```
plt.legend()
plt.show()

# Example single patient (use columns order from original X)
example_patient = {
  # Provide realistic example values — adjust as needed
  'age': 56, 'sex': 1, 'cp': 2, 'trestbps': 130, 'chol': 236, 'fbs': 0,
  'restecg': 1, 'thalach': 178, 'exang': 0, 'oldpeak': 0.8, 'slope': 2, 'ca': 0, 'thal': 2
}

# Create DataFrame row in same column order
example_df = pd.DataFrame([example_patient])[X.columns]
print("Example patient row:")
display(example_df)

# Transform using preprocessor and predict with best model
example_X = preprocessor.transform(example_df)
pred_rf = best_rf.predict(example_X)[0]
prob_rf = best_rf.predict_proba(example_X)[:,1][0]
print(f"RandomForest prediction: {pred_rf} (1 = disease, 0 = no disease). Probability: {prob_rf:.3f}")

pred_lr = lr.predict(example_X)[0]
prob_lr = lr.predict_proba(example_X)[:,1][0]
print(f"LogisticRegression prediction: {pred_lr} Probability: {prob_lr:.3f}")

pred_nn = (np.ravel(nn_model.predict(example_X)) > 0.5).astype(int)[0]
prob_nn = np.ravel(nn_model.predict(example_X))[0]
print(f"NeuralNetwork prediction: {pred_nn} Probability: {prob_nn:.3f}")
```

```
model_dir = "models"
os.makedirs(model_dir, exist_ok=True)

joblib.dump(best_rf, os.path.join(model_dir, "best_random_forest.joblib"))
print("Saved best_random_forest.joblib")
joblib.dump(lr, os.path.join(model_dir, "logistic_regression.joblib"))
print("Saved logistic_regression.joblib")
joblib.dump(preprocessor, os.path.join(model_dir, "preprocessor.joblib"))
print("Saved preprocessor.joblib")
nn_model.save(os.path.join(model_dir, "nn_model.h5"))
print("Saved nn_model.h5")

print(f"All artifacts saved in folder: {model_dir} (download from Colab Files panel).")

print("PROJECT SUMMARY — Student-style conclusions:\n")
print("1) Problem: Predict heart disease (binary) using clinical features.")
print("2) Data: heart.csv loaded. Checked data, no major missing values found.")
print("3) EDA: Explored distributions, categorical relationships, and correlations.")
print("4) Preprocessing: One-hot encoded cp, thal, slope; scaled numeric features with StandardScaler.")
print("5) Models trained: Logistic Regression, Decision Tree, Random Forest (base + tuned), Neural Network.")
print("6) Tuning: RandomizedSearchCV performed for RandomForest.")
print("7) Evaluation: Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrices, ROC curves.")
print("8) Interpretation: Feature importances from RandomForest and coefficients from LogisticRegression shown.")
print("9) Example inference: Showed how to predict for a new single patient using the preprocessor + model.")
print("10) Artifacts: Saved models and preprocessor for later inference or deployment.")
print("\nLimitations to mention in your report:")
print("- Dataset size (~303 rows) — risk of overfitting and limited generalization.")
```

```
print("- No external cohort validation included.")

print("- Preprocessing choices (which columns to one-hot) were decided for simplicity; alternative encodings possible.")

print("\nNext steps (if you want): add SHAP explanations, nested CV, class imbalance methods (SMOTE), or deployment. Tell me which and I'll add it.")


# ==========================================

# HEART DISEASE PREDICTION – GRADIO APP (HF Spaces Ready)

# ==========================================


!pip install gradio joblib -q


import gradio as gr

import pandas as pd

import joblib

from sklearn.pipeline import Pipeline


# Re-load the preprocessor and the best_rf model

preprocessor = joblib.load("models/preprocessor.joblib")

best_rf = joblib.load("models/best_random_forest.joblib")


# Create a pipeline that includes preprocessing and the model

# The ColumnTransformer must be part of the pipeline to process raw inputs

model_pipeline = Pipeline(steps=[

    ('preprocessor', preprocessor),

    ('classifier', best_rf)

])


# Save the pipeline so Gradio can load it by name
```

```
joblib.dump(model_pipeline, "models/best_random_forest_pipeline.joblib")
print("Saved models/best_random_forest_pipeline.joblib")

# Load saved RF pipeline
model = joblib.load("models/best_random_forest_pipeline.joblib")

def predict_heart_disease(age, sex, cp, trestbps, chol, fbs, restecg,
                          thalach, exang, oldpeak, slope, ca, thal):

    # Create input dataframe - ensure column order matches original X
    input_data = pd.DataFrame([{
        "age": age,
        "sex": sex,
        "cp": cp,
        "trestbps": trestbps,
        "chol": chol,
        "fbs": fbs,
        "restecg": restecg,
        "thalach": thalach,
        "exang": exang,
        "oldpeak": oldpeak,
        "slope": slope,
        "ca": ca,
        "thal": thal
    }])

    # Predict using saved pipeline (preprocessing is handled internally by the pipeline)
    proba = model.predict_proba(input_data)[0][1]
    label = int(proba >= 0.5)
```

```
    return {
        "Probability of Heart Disease": float(proba),
        "Prediction (0=No, 1=Yes)": label
    }

# Gradio UI - Simplified inputs to directly take numerical values
inputs = [
    gr.Number(label="Age"),
    gr.Radio([0, 1], label="Sex (0=Female, 1=Male)"),
    gr.Radio([0, 1, 2, 3], label="Chest Pain Type (cp)"),
    gr.Number(label="Resting BP (trestbps)"),
    gr.Number(label="Cholesterol (chol)"),
    gr.Radio([0, 1], label="Fasting Blood Sugar >120 mg/dl (0=False, 1=True)"),
    gr.Radio([0, 1, 2], label="Resting ECG (restecg)"),
    gr.Number(label="Max Heart Rate (thalach)"),
    gr.Radio([0, 1], label="Exercise Induced Angina (0=No, 1=Yes)"),
    gr.Number(label="ST Depression (oldpeak)"),
    gr.Radio([0, 1, 2], label="Slope of Peak Exercise ST Segment (slope)"),
    gr.Radio([0, 1, 2, 3, 4], label="Number of Major Vessels (ca)"),
    gr.Radio([0, 1, 2, 3], label="Thal (0,1=Normal, 2=Fixed Defect, 3=Reversible Defect)") # Correcting thal input based on EDA (0,1,2,3)
]

output = gr.JSON()

gr.Interface(
    fn=predict_heart_disease,
    inputs=inputs,
```

```
    outputs=output,
    title="Heart Disease Prediction App",
    description="A Machine Learning model predicting the probability of heart disease.",
    allow_flagging="never"
).launch(share=True)
```

OUTPUTS

Imports complete.

Dataset loaded. Shape: (303, 14)

First 5 rows:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

Data types:

age int64

sex int64

cp int64

trestbps int64

chol int64

fbs int64

restecg int64

thalach int64

exang int64

oldpeak float64

slope int64

ca int64

thal int64

target int64

dtype: object

Missing values per column:

age 0

sex 0

cp 0

trestbps 0

chol 0

fbs 0

restecg 0

thalach 0

exang 0

oldpeak 0

slope 0

ca 0

thal 0

target 0

dtype: int64

Column names and quick meanings:

- age
- sex
- cp
- trestbps
- chol
- fbs
- restecg
- thalach
- exang
- oldpeak
- slope
- ca
- thal

- target

Target value counts (0 = no disease, 1 = disease):

target

1 165

0 138

Name: count, dtype: int64

Summary statistics (numeric):

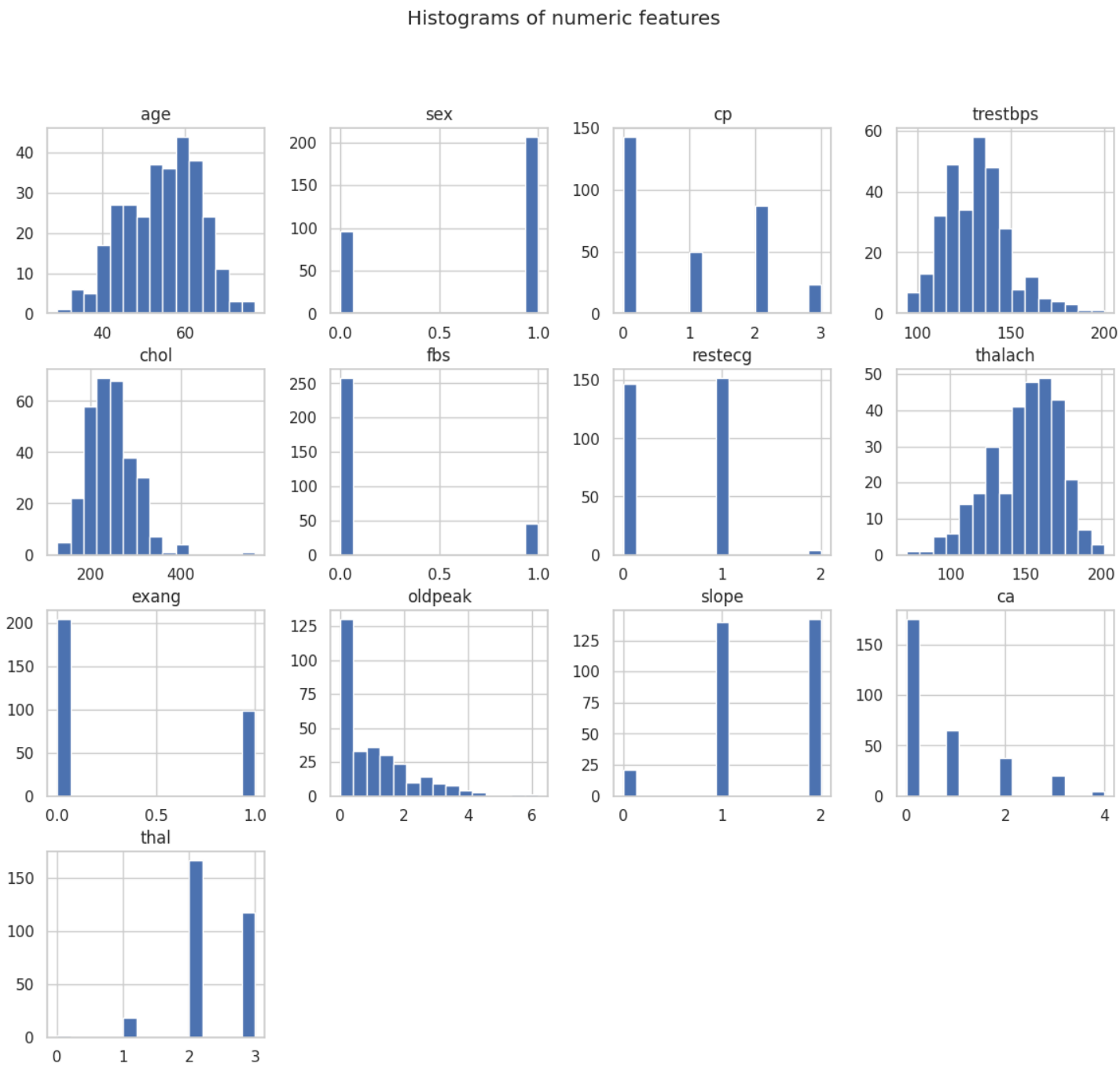
	count	mean	std	min	25%	50%	75%	max
age	303.0	54.366337	9.082101	29.0	47.5	55.0	61.0	77.0
sex	303.0	0.683168	0.466011	0.0	0.0	1.0	1.0	1.0
cp	303.0	0.966997	1.032052	0.0	0.0	1.0	2.0	3.0
trestbps	303.0	131.623762	17.538143	94.0	120.0	130.0	140.0	200.0
chol	303.0	246.264026	51.830751	126.0	211.0	240.0	274.5	564.0
fbs	303.0	0.148515	0.356198	0.0	0.0	0.0	0.0	1.0
restecg	303.0	0.528053	0.525860	0.0	0.0	1.0	1.0	2.0
thalach	303.0	149.646865	22.905161	71.0	133.5	153.0	166.0	202.0
exang	303.0	0.326733	0.469794	0.0	0.0	0.0	1.0	1.0
oldpeak	303.0	1.039604	1.161075	0.0	0.0	0.8	1.6	6.2
slope	303.0	1.399340	0.616226	0.0	1.0	1.0	2.0	2.0
ca	303.0	0.729373	1.022606	0.0	0.0	0.0	1.0	4.0
thal	303.0	2.313531	0.612277	0.0	2.0	2.0	3.0	3.0
target	303.0	0.544554	0.498835	0.0	0.0	1.0	1.0	1.0

Observations to note:

- Check min/max for suspicious values (e.g., 'thal' or 'ca' sometimes encoded with 0 or missing).

- Dataset rows: 303

<Figure size 1400x1000 with 0 Axes>



Categorical-like columns used for plotting: ['sex', 'cp', 'fbs', 'restecg', 'exang', 'slope', 'ca', 'thal']

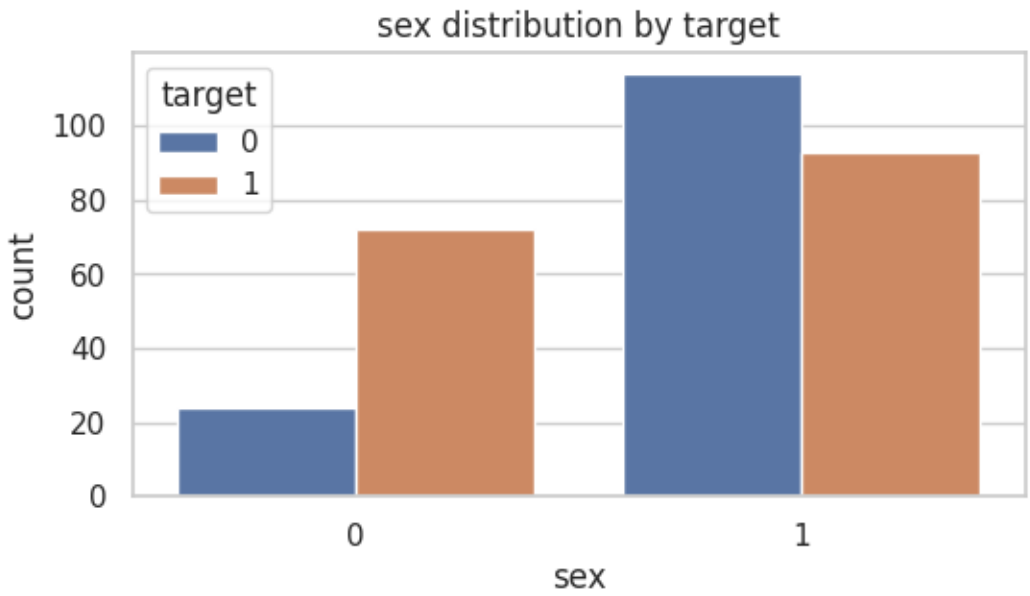
Value counts for sex:

sex

1 207

0 96

Name: count, dtype: int64



Value counts for cp:

cp

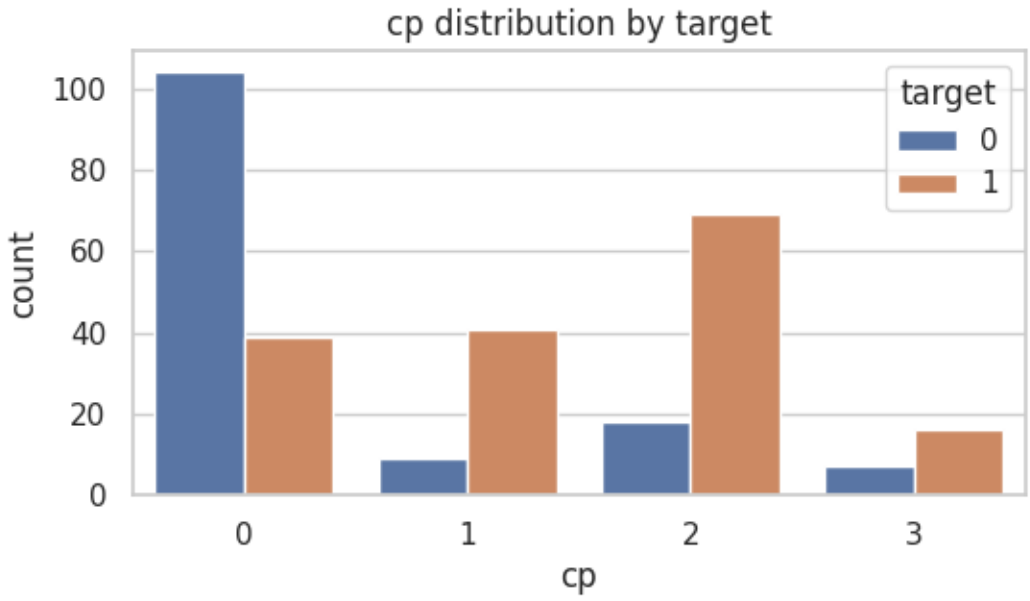
0 143

2 87

1 50

3 23

Name: count, dtype: int64



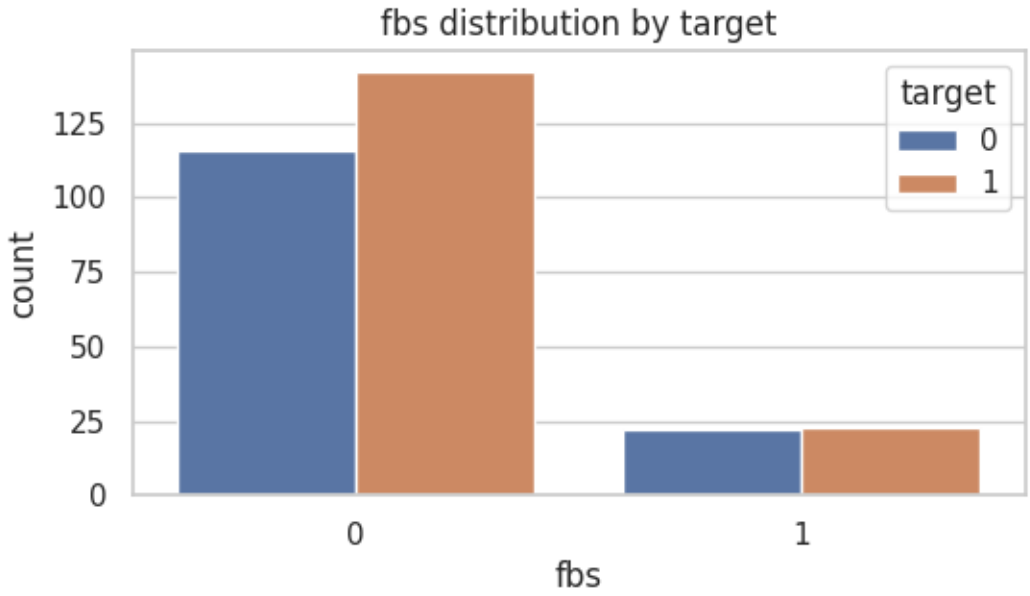
Value counts for fbs:

fbs

0 258

1 45

Name: count, dtype: int64



Value counts for restecg:

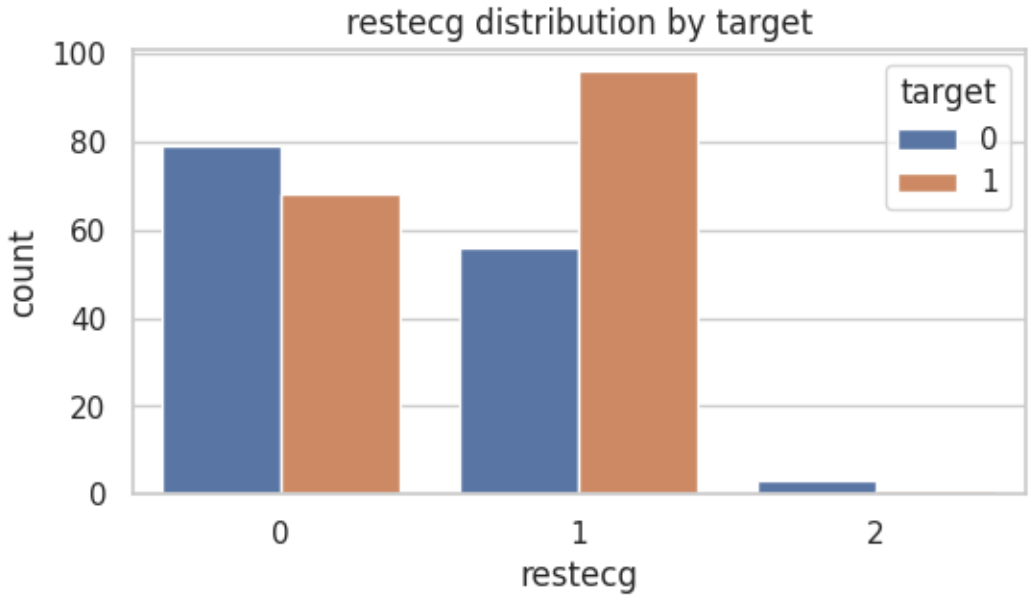
restecg

1 152

0 147

2 4

Name: count, dtype: int64



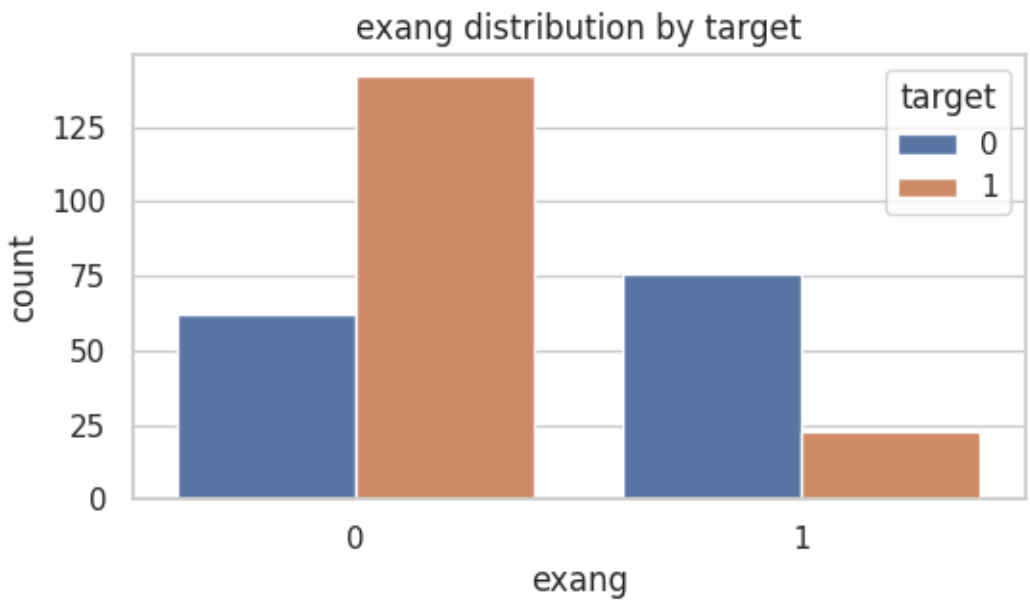
Value counts for exang:

exang

0 204

1 99

Name: count, dtype: int64



Value counts for slope:

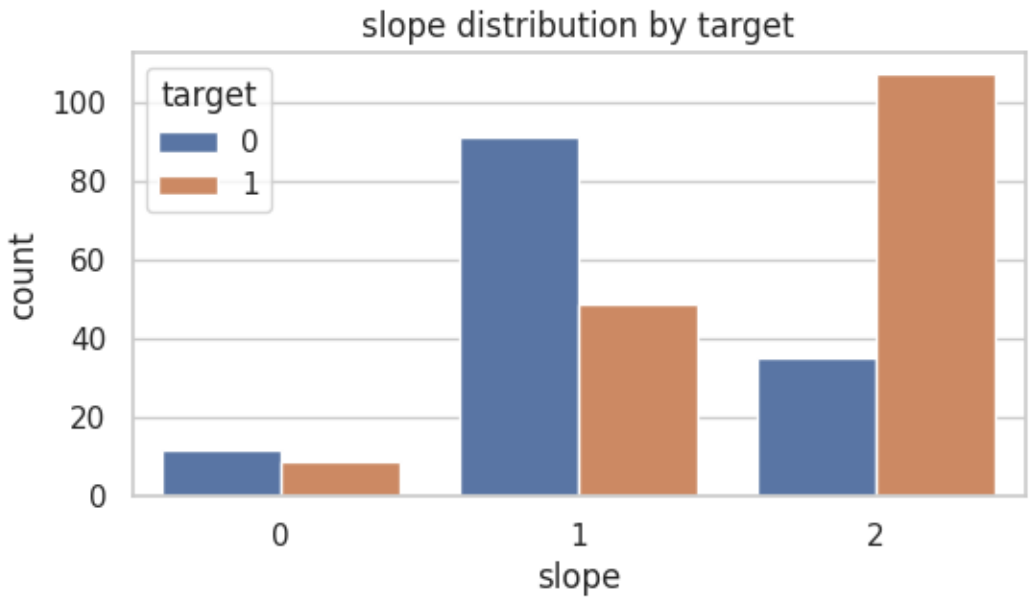
slope

2 142

1 140

0 21

Name: count, dtype: int64



Value counts for ca:

ca

0 175

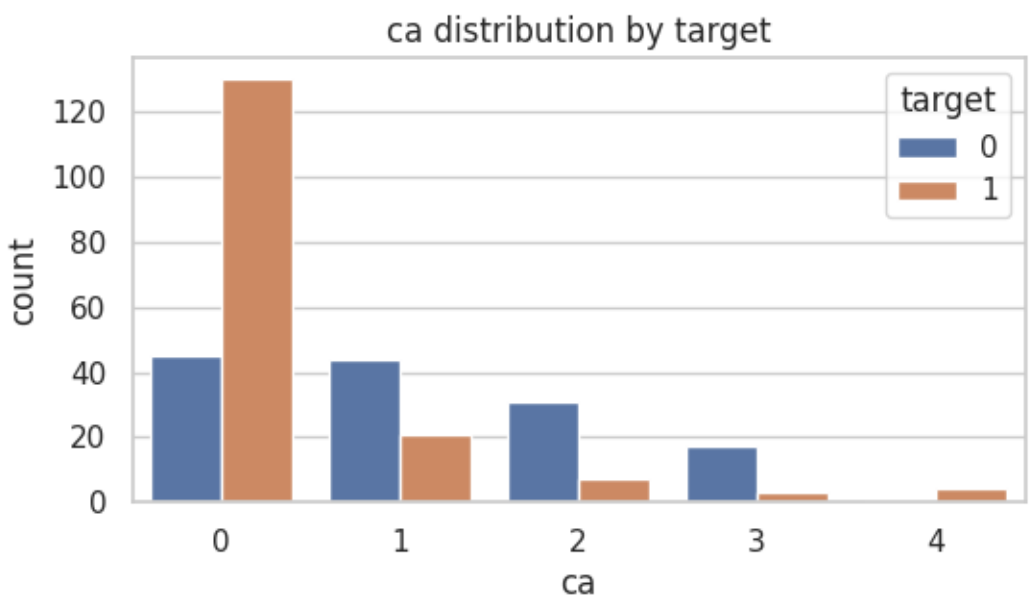
1 65

2 38

3 20

4 5

Name: count, dtype: int64



Value counts for thal:

thal

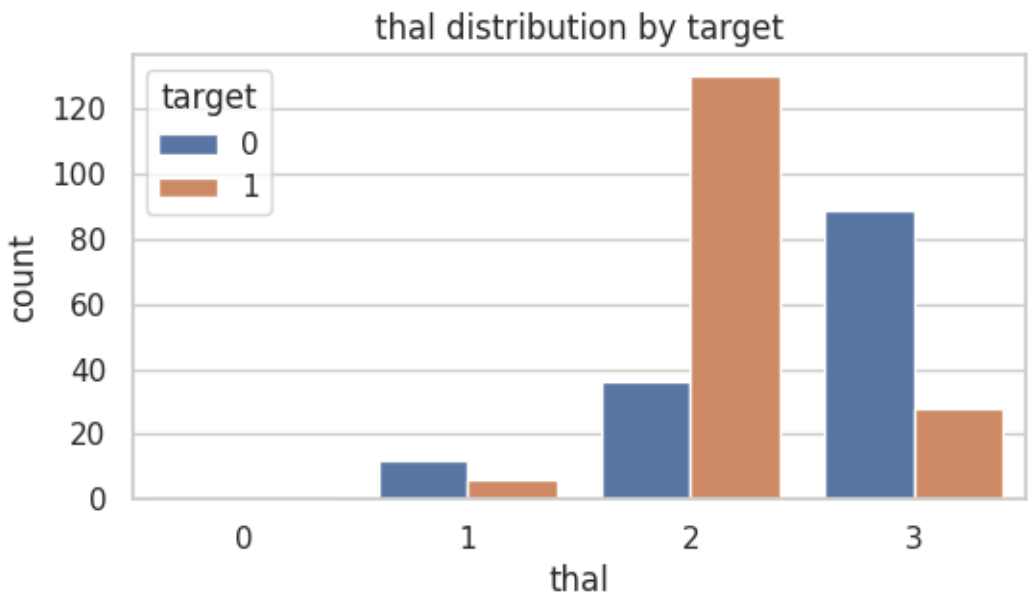
2 166

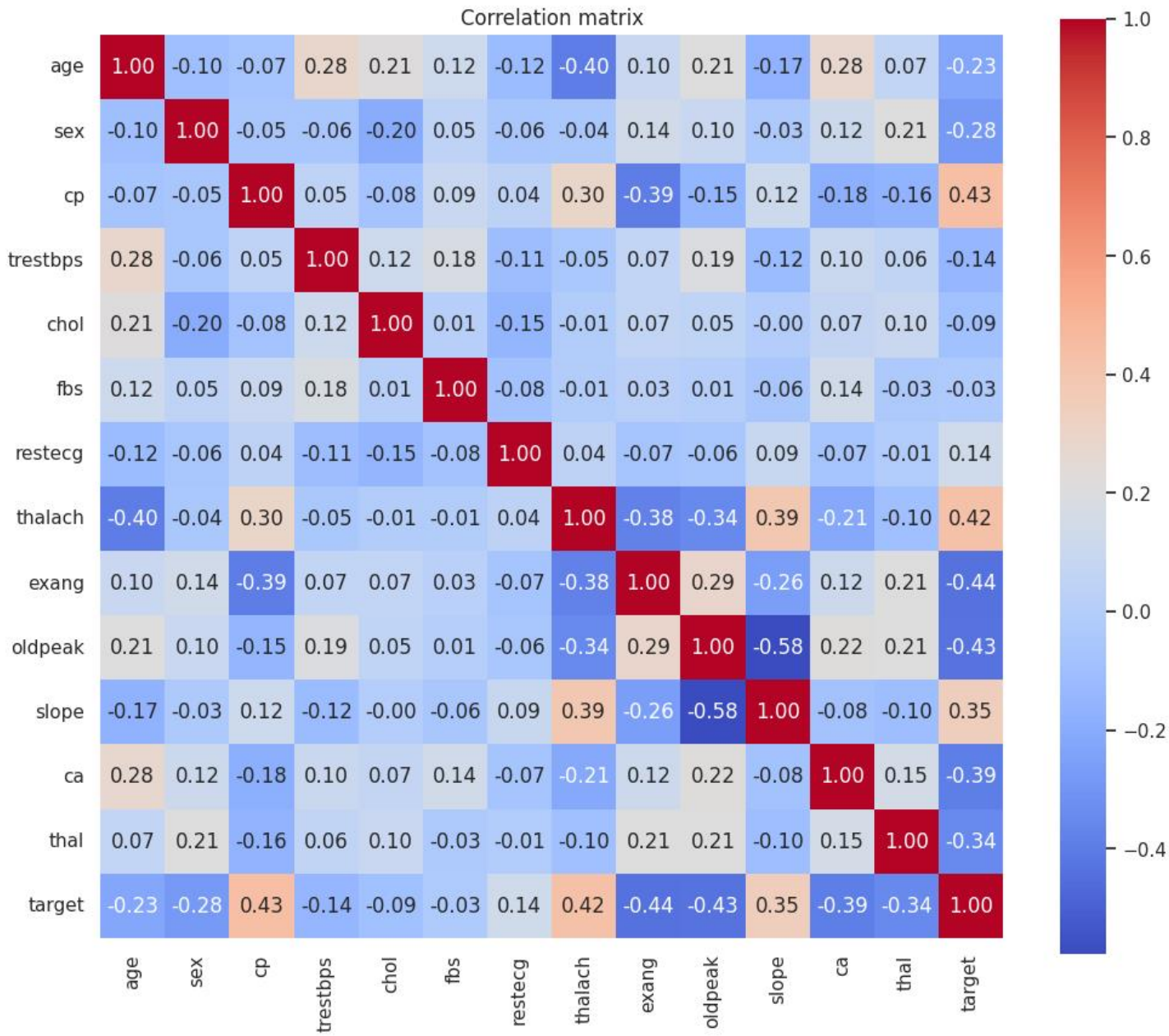
3 117

1 18

0 2

Name: count, dtype: int64





Look for features strongly correlated with 'target' and potential multicollinearity.

Checking for any non-numeric or unusual encodings:

Check for zeros or impossible values in columns that should be positive (like 'thal' sometimes 0):

thal unique values: [np.int64(0), np.int64(1), np.int64(2), np.int64(3)]

ca unique values: [np.int64(0), np.int64(1), np.int64(2), np.int64(3), np.int64(4)]

If any '?' or string placeholders are present, we should replace them and convert columns to numeric. This dataset looks numeric-clean.

One-hot columns: ['cp', 'thal', 'slope']

Numerical columns: ['age', 'sex', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'ca']

Preprocessing done. Transformed shape: (303, 18)

Sample feature names (first 20): ['age', 'sex', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'ca', 'cp_1', 'cp_2', 'cp_3', 'thal_1', 'thal_2', 'thal_3', 'slope_1', 'slope_2']

Train shape: (242, 18) Test shape: (61, 18)

Train target distribution:

[110 132]

Test target distribution:

[28 33]

Training Logistic Regression...

>>> Evaluating: Logistic Regression

Accuracy: 0.8033

Precision: 0.7838

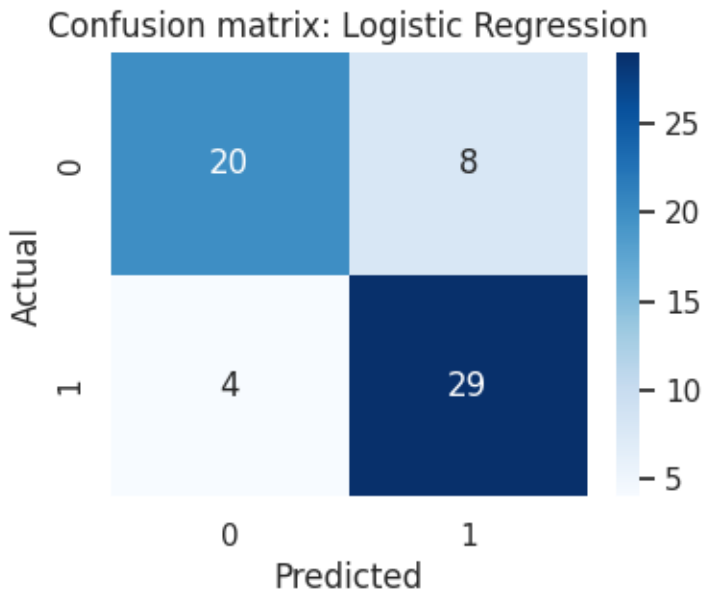
Recall: 0.8788

F1-score: 0.8286

ROC-AUC: 0.8885

Classification report:

	precision	recall	f1-score	support
0	0.83	0.71	0.77	28
1	0.78	0.88	0.83	33
accuracy			0.80	61
macro avg	0.81	0.80	0.80	61
weighted avg	0.81	0.80	0.80	61



Training Decision Tree...

>>> Evaluating: Decision Tree

Accuracy: 0.7213

Precision: 0.7222

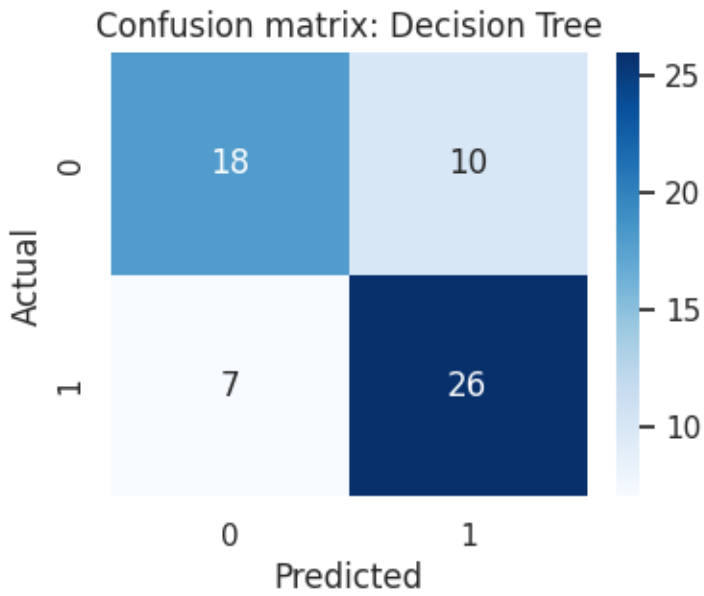
Recall: 0.7879

F1-score: 0.7536

ROC-AUC: 0.7154

Classification report:

	precision	recall	f1-score	support
0	0.72	0.64	0.68	28
1	0.72	0.79	0.75	33
accuracy			0.72	61
macro avg	0.72	0.72	0.72	61
weighted avg	0.72	0.72	0.72	61



Training Random Forest (100 trees)...

>>> Evaluating: Random Forest (base)

Accuracy: 0.7705

Precision: 0.7436

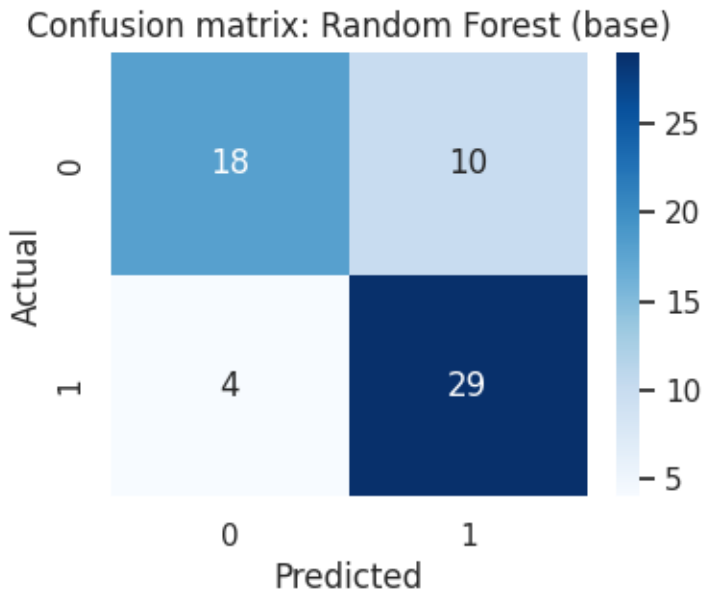
Recall: 0.8788

F1-score: 0.8056

ROC-AUC: 0.9134

Classification report:

	precision	recall	f1-score	support
0	0.82	0.64	0.72	28
1	0.74	0.88	0.81	33
accuracy			0.77	61
macro avg	0.78	0.76	0.76	61
weighted avg	0.78	0.77	0.77	61



Hyperparameter tuning (RandomizedSearchCV) for RandomForest — may take a few minutes.

Fitting 5 folds for each of 20 candidates, totalling 100 fits

Best params: {'n_estimators': 200, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_features': 'log2', 'max_depth': 20}

>>> Evaluating: Random Forest (tuned)

Accuracy: 0.7705

Precision: 0.7436

Recall: 0.8788

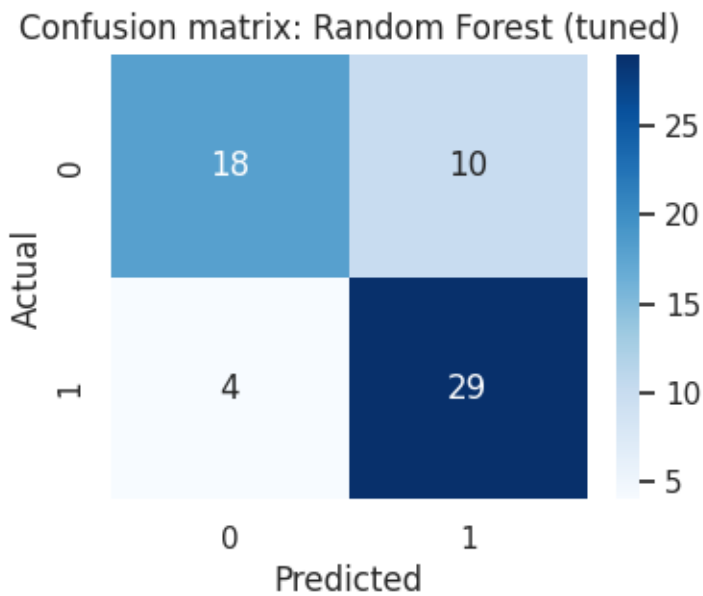
F1-score: 0.8056

ROC-AUC: 0.9102

Classification report:

	precision	recall	f1-score	support
0	0.82	0.64	0.72	28
1	0.74	0.88	0.81	33
accuracy			0.77	61

macro avg	**0.78**	**0.76**	**0.76**	**61**
weighted avg	**0.78**	**0.77**	**0.77**	**61**



Training a simple feedforward neural network (Keras).

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dense_3 (Dense)	(None, 64)	1,216
dropout_2 (Dropout)	(None, 64)	0
dense_4 (Dense)	(None, 32)	2,080
dropout_3 (Dropout)	(None, 32)	0
dense_5 (Dense)	(None, 1)	33

```
└──────────────────────────────────┴──────────────────────────────────┴──────────────────────────┘

Total params: 3,329 (13.00 KB)

Trainable params: 3,329 (13.00 KB)

Non-trainable params: 0 (0.00 B)

Epoch 1/200

13/13 ━━━
```

```
Epoch 10/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8226 - loss: 0.3666 -
val_accuracy: 0.8776 - val_loss: 0.2940
Epoch 11/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.8444 - loss: 0.3557 -
val_accuracy: 0.8980 - val_loss: 0.2904
Epoch 12/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.8091 - loss: 0.4118 -
val_accuracy: 0.8571 - val_loss: 0.2946
Epoch 13/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8423 - loss: 0.3435 -
val_accuracy: 0.8571 - val_loss: 0.2966
Epoch 14/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8641 - loss: 0.3631 -
val_accuracy: 0.8367 - val_loss: 0.2938
Epoch 15/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8659 - loss: 0.3036 -
val_accuracy: 0.8367 - val_loss: 0.2933
Epoch 16/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.8883 - loss: 0.3165 -
val_accuracy: 0.8571 - val_loss: 0.2949
Epoch 17/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8662 - loss: 0.3082 -
val_accuracy: 0.8571 - val_loss: 0.2943
Epoch 18/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.8490 - loss: 0.3162 -
val_accuracy: 0.8367 - val_loss: 0.2984
Epoch 19/200
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.8706 - loss: 0.2891 -
val_accuracy: 0.8367 - val_loss: 0.3012
Epoch 20/200
```

13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8995 - loss: 0.2722 - val_accuracy: 0.8367 - val_loss: 0.3024

Epoch 21/200

13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8640 - loss: 0.3000 - val_accuracy: 0.8367 - val_loss: 0.3078

Epoch 22/200

13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.8387 - loss: 0.3455 - val_accuracy: 0.8163 - val_loss: 0.3260

Epoch 23/200

13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9044 - loss: 0.2719 - val_accuracy: 0.8163 - val_loss: 0.3095

Epoch 24/200

13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8743 - loss: 0.3359 - val_accuracy: 0.8163 - val_loss: 0.3086

Epoch 25/200

13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8808 - loss: 0.2835 - val_accuracy: 0.8163 - val_loss: 0.3085

Epoch 26/200

13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9086 - loss: 0.2426 - val_accuracy: 0.8163 - val_loss: 0.3174

Epoch 26: early stopping

Restoring model weights from the end of the best epoch: 11.

>>> Evaluating: Neural Network

2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 63ms/step

Accuracy: 0.8033

Precision: 0.7838

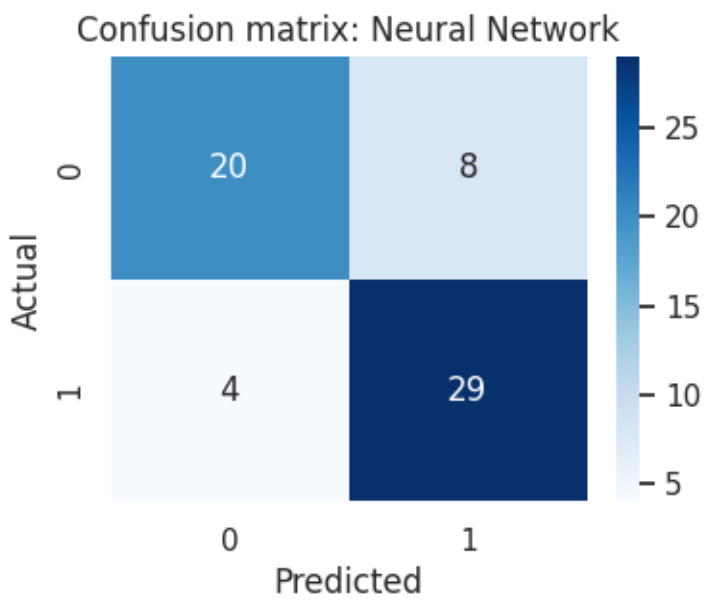
Recall: 0.8788

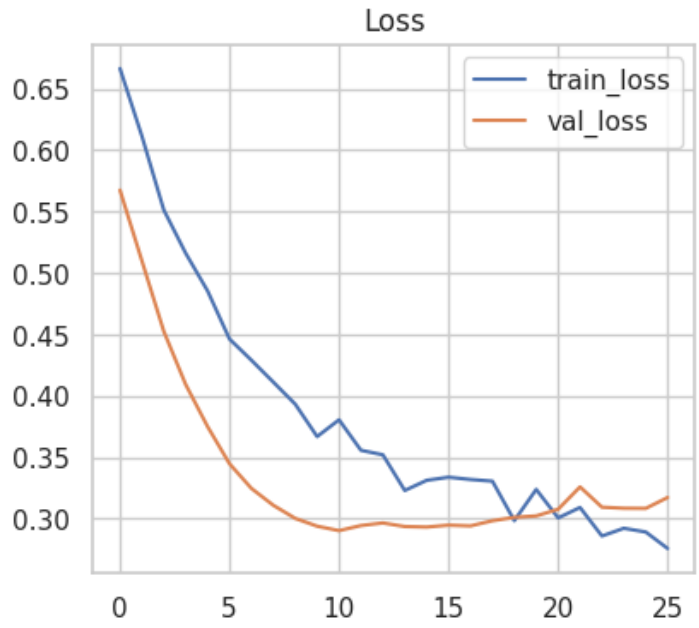
F1-score: 0.8286

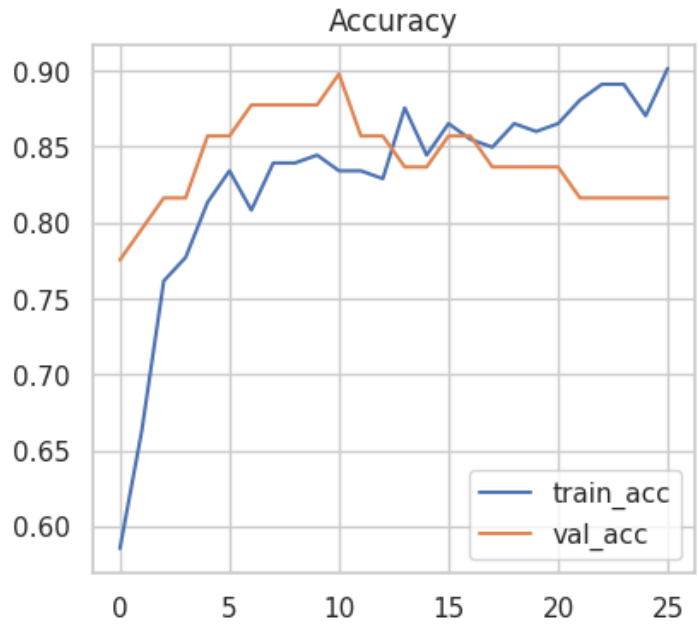
ROC-AUC: 0.8745

Classification report:

	precision	recall	f1-score	support
0	0.83	0.71	0.77	28
1	0.78	0.88	0.83	33
accuracy			0.80	61
macro avg	0.81	0.80	0.80	61
weighted avg	0.81	0.80	0.80	61







Model comparison (sorted by ROC-AUC):

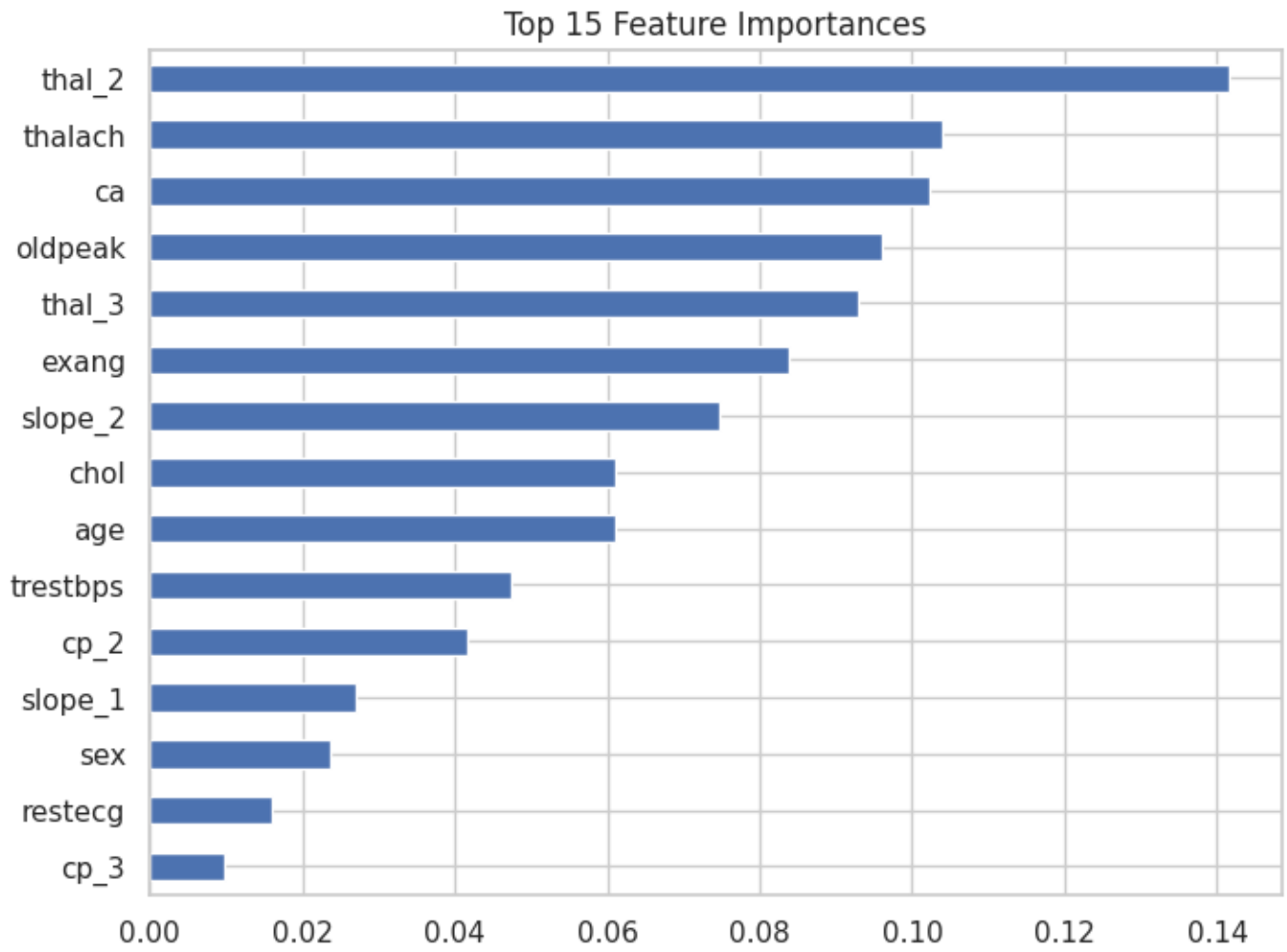
	model	accuracy	precision	recall	f1	roc_auc
2	Random Forest (base)	0.7705	0.7436	0.8788	0.8056	0.9134
3	Random Forest (tuned)	0.7705	0.7436	0.8788	0.8056	0.9102
0	Logistic Regression	0.8033	0.7838	0.8788	0.8286	0.8885
4	Neural Network	0.8033	0.7838	0.8788	0.8286	0.8745
1	Decision Tree	0.7213	0.7222	0.7879	0.7536	0.7154

Top 15 features by RandomForest importance:

	0
thal_2	0.141509
thalach	0.103923
ca	0.102263
oldpeak	0.096030
thal_3	0.092974
exang	0.083771

	0
slope_2	0.074685
chol	0.060997
age	0.060950
trestbps	0.047393
cp_2	0.041645
slope_1	0.026964
sex	0.023785
restecg	0.015948
cp_3	0.009885

dtype: float64



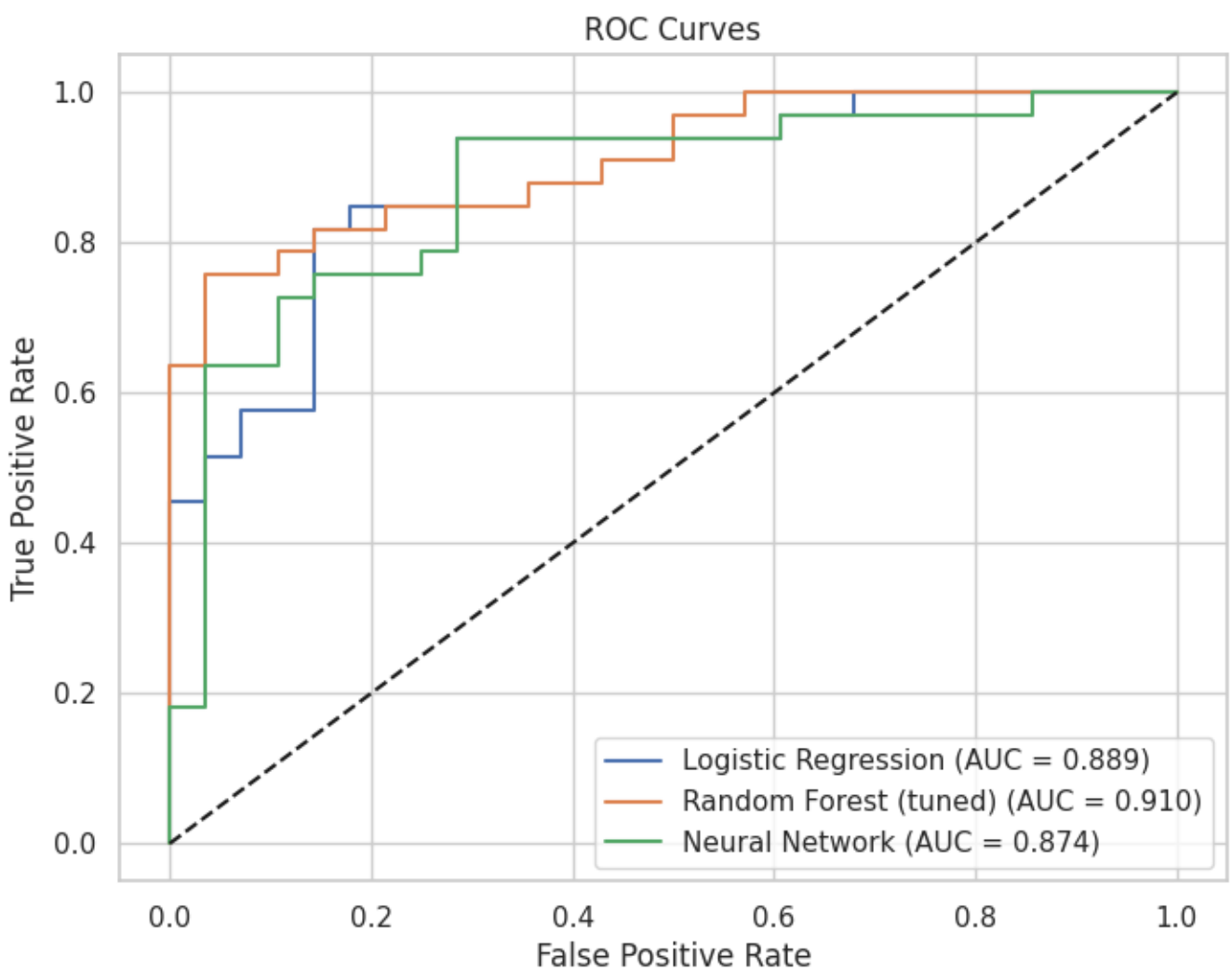
Top features by absolute LogisticRegression coefficient:

	0
cp_2	**1.435292**
cp_3	**1.279855**
thal_3	**-0.845484**
thal_2	**0.748302**
cp_1	**0.732775**
chol	**-0.567438**
ca	**-0.540689**
sex	**-0.510295**
exang	**-0.468281**

	0
oldpeak	-0.440787
thalach	0.379160
slope_1	-0.364167
slope_2	0.281476
restecg	0.214190
trestbps	-0.148547

dtype: float64

2/2 ━━ 0s 41ms/step



Example patient row:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	56	1	2	130	236	0	1	178	0	0.8	2	0	2

RandomForest prediction: 1 (1 = disease, 0 = no disease). Probability: 0.966

LogisticRegression prediction: 1 Probability: 0.964

1/1 ━━ 0s 57ms/step

1/1 ━━ 0s 48ms/step

NeuralNetwork prediction: 1 Probability: 0.942

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`.

Saved best_random_forest.joblib

Saved logistic_regression.joblib

Saved preprocessor.joblib

Saved nn_model.h5

All artifacts saved in folder: models (download from Colab Files panel).

PROJECT SUMMARY — Student-style conclusions:

1) Problem: Predict heart disease (binary) using clinical features.

2) Data: heart.csv loaded. Checked data, no major missing values found.

3) EDA: Explored distributions, categorical relationships, and correlations.

4) Preprocessing: One-hot encoded cp, thal, slope; scaled numeric features with StandardScaler.

5) Models trained: Logistic Regression, Decision Tree, Random Forest (base + tuned), Neural Network.

6) Tuning: RandomizedSearchCV performed for RandomForest.

7) Evaluation: Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrices, ROC curves.

8) Interpretation: Feature importances from RandomForest and coefficients from LogisticRegression shown.

9) Example inference: Showed how to predict for a new single patient using the preprocessor + model.

10) Artifacts: Saved models and preprocessor for later inference or deployment.

Limitations to mention in your report:

- Dataset size (~303 rows) — risk of overfitting and limited generalization.

- No external cohort validation included.

- Preprocessing choices (which columns to one-hot) were decided for simplicity; alternative encodings possible.

Next steps (if you want): add SHAP explanations, nested CV, class imbalance methods (SMOTE), or deployment. Tell me which and I'll add it.

Saved models/best_random_forest_pipeline.joblib

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

*** Running on public URL:**

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory to deploy to Hugging Face Spaces (https://huggingface.co/spaces)